

SMS Excavator with CMS model

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PROJECT NAME:

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CLIENT NAME:

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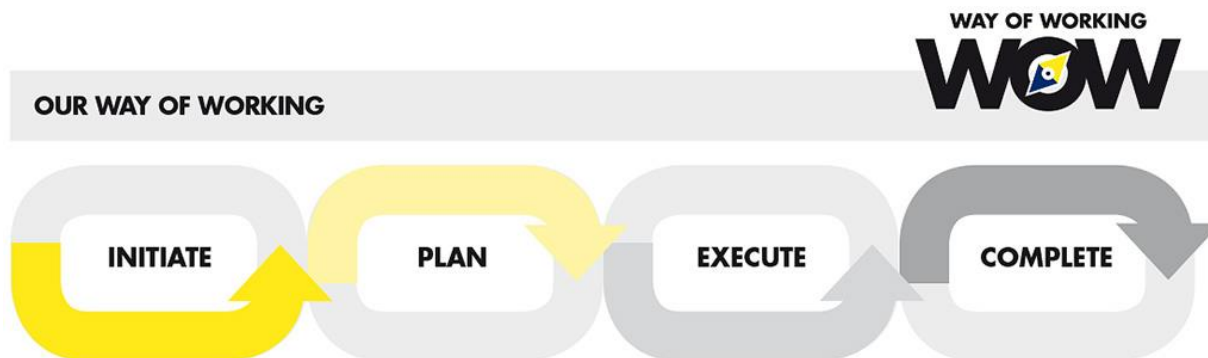
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1. INTRODUCTION

This document forms part of the Boskalis Way of Working, the integrated quality management system applicable to all operations in Boskalis. The Boskalis Way of Working is structured around four Phases as pictured below. This SMS CMS Excavator is typically prepared in the Plan Phase, where the main implementation is taking place in the Execute Phase.



More detailed information about the Boskalis Way of Working can be found in the Group Manual and the User Guide. A dedicated website with all supporting materials is available at wow.boskalis.com.

1.1. Scope of Document

The purpose of this document is to describe the Crane Monitoring System (CMS) on excavators and the survey method to be followed for surveys performed with the CMS. To ensure that the CMS is setup properly in relation to setup, geometry and calibrations. And surveys carried out with the CMS are within the specifications as set in this document.

2. SENSORS AND SYSTEMS

<<Add or remove as applicable for planned installation>>

The following types of equipment are installed as part of the CMS:

Ref	Description	Objective	Type
1	RTK GNSS receiver	Excavator position	Trimble SPS85x
2	Heading sensor	Excavator heading	Trimble SPS55xH
3	Pitch / Roll sensor	Excavator attitude	AE sensors XB2i
4	Angler sensors	Boom, Stick and bucket angles	Seatools Rotax HD Digital
5	Computer		NISE with DV CMS software
6	Connection box	Interfacing & Power	

2.1. RTK DGNSS Positioning

For the excavator positioning a Trimble SPS85x RTK DGNSS system will be installed including antenna. The Trimble SPS85x receiver will receive differential corrections from a Trimble SPS85x RTK-Base installed near the project location. The RTK Corrections will provide a position accuracy of < 10 mm horizontal and < 20 mm vertical and are capable to provide position update rates of at least every 1 second with RTK corrections not exceeding 8 seconds.

2.2. DGNSS Heading

For the excavator heading a Trimble SPS55x GNSS Heading option will be installed including antenna. The SPS55xH will be interfaced with the SPS85x positioning system and with the positions calculated from both systems an accurate heading will be determined. The heading accuracy of the SPS55xH is better than 0.10° RMS with a 2 meter antenna separation.

2.3. Roll & pitch

For the roll and pitch of the excavator an AE sensors XB2i will be installed. The XB2i is installed in a watertight housing (IP65) and has a range of +/- 10°. The XB2i roll and pitch sensor has an accuracy better than 0.10°.

2.4. Anglers

For the determination of the angles of Boom, Stick and bucket and Jib if applicable for each angle a Seatools Rotax HD Digital angler will be installed. The Seatools Rotax HD Digital angler sensor is a heavy duty sensor for use underwater and designed to be used on Backhoe dredgers or excavators. The sensor has an accuracy of better 0.10% full scale (< 0.3°).

3. SOFTWARE

<<Add or remove as applicable for planned installation>>

The software used is the DredgeView Crane Monitoring System software. The software is in-house developed by Boskalis.

3.1. General description

DV-CMS is the DredgeView 2.0 Excavator, Backhoe and Grab dredger navigation and production module custom tuned to each Boskalis Backhoe, Excavator or Grab dredger. DV-CMS integrates precise 3D positioning, dredger modelling, project design information and survey results into real time graphical displays for both crane and barge operators. These displays allow the operator to dredge or build accurately to the required design. The position of the grab or bucket can be logged in readable CSV format at time interval, at deepest point of excavation, or in case of backfill at the position of each individual dump.

3.2. Survey functionality

The DV-CMS software can also be used to perform surveys, the position of the reference point of the bucket can be logged in a comma delimited text file with position and height of the reference point. Surveys with excavators are done when other survey methods are not available due to safety reasons.

4. MOBILIZATION, INSTALLATION AND CALIBRATION

<<Add or remove as applicable for planned installation>>

4.1. Mobilization and installation

The mobilization and installation of a DredgeView Crane Monitoring System on a shore based excavator is coordinated by CONTRACTOR.

Installation of the DV CMS will be performed by an electronic engineer from CONTRACTOR with support from Project.

4.2. Geometry

4.2.1. Geometry definition

Origin of the grid in the DV CMS system for the excavator is the kingpin for the X and Y value and the bottom of tracks for the Z value.

The Y axis is along the boom and stick positive forwards and the X-axis is positive to starboard perpendicular to the Y-axis. The Z-axis is positive upwards perpendicular to the X-Y plane.

The roll is positive when the crane rotates around the Y-axis to starboard, the pitch is positive when the crane rotates around the X-axis, backwards.

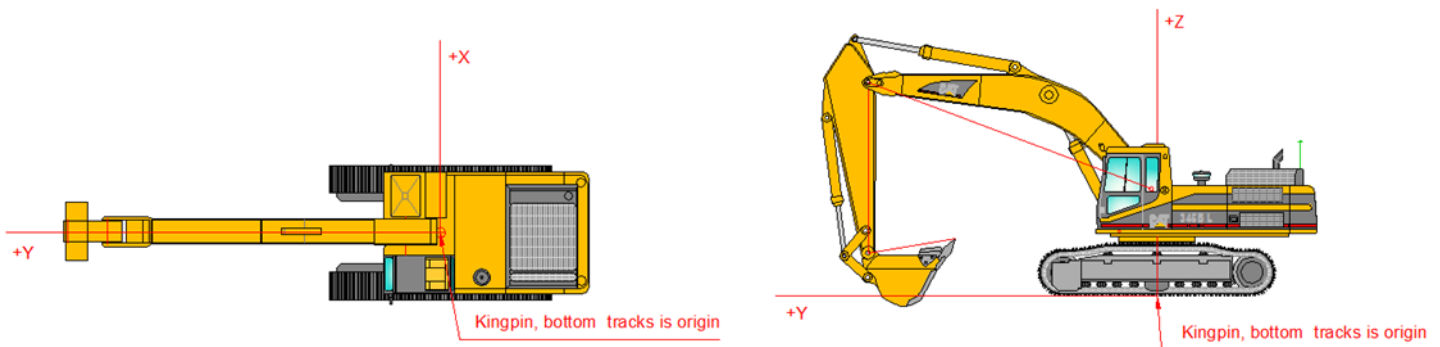


Figure 1: Origin and axis

4.2.2. Locations and dimensions

<<Add or remove as applicable for planned installation>>

The geometry of the following items will be determined with land survey techniques:

Ref	Dimension	Definition
0	Kingpin, bottom of tracks	Origin of grid
1	Main antenna	RTK GNSS antenna reference point
2	Trunnion	Rotation axis of the boom
3	Boom length	Distance Boom pin – Stick Pin
4	Stick length	Distance Stick Pin – Bucket Pin
5	Bucket length	Distance Bucket Pin – Bucket knife/teeth reference point

6	Bucket width	Distance between the sides of the bucket at the blade.
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4.3. Calibrations

<<Describe here how the calibrations will or have been performed for all survey systems where applicable. Add or remove calibration procedures as required for the project.>>

Prior to shipment to the project all survey equipment will be tested for good working order. Where applicable the factory calibration will be checked if still valid and in accordance with Boskalis procedure SURV-200.

On the project the survey equipment will be tested and calibrated in accordance with 'SURV-200 Survey Field checks and equipment test sheets' where applicable.

The following calibration procedures will be used:

Boskalis Documents	
Document No.	Document Title
SURV-205	Field Check for DGNS receiver
SURV-207	Field Check for gyro compass
SURV-209	Field Check for motion sensor
SURV-220	Field Check for excavator calibration RTK
SURV-221	Field check Position check CMS

Project Documents	
Document No.	Document Title

4.4. Tolerances

<<Adjust add or remove as required for the project. Be aware that from the Field check procedures is referred to this section.>>

When performing Calibrations and verifications the following tolerances will be allowed for:

Ref	Dimension	Definition
1	RTK GNSS	< 0.02 m (2 σ) Horizontal, < 0.03 (2 σ) Vertical.
2	GNSS Heading	< 0.2°
3	Roll and Pitch	< 0.2°
4	Geometry	< 0.005 m
5	Position check measuring point (Bucket)	< 0.15 Horizontal, < 0.10 m Vertical

4.5. Error budget

<<Add, remove or modify as applicable for the planned installation>>

Every measuring device works to a pre-specified accuracy of the device. There are 3 error components:

- Vertical error;
- Horizontal error which can be divided in two components:
 - Error parallel to the crane (and boom – stick);
 - Error perpendicular to the crane (and boom – stick).

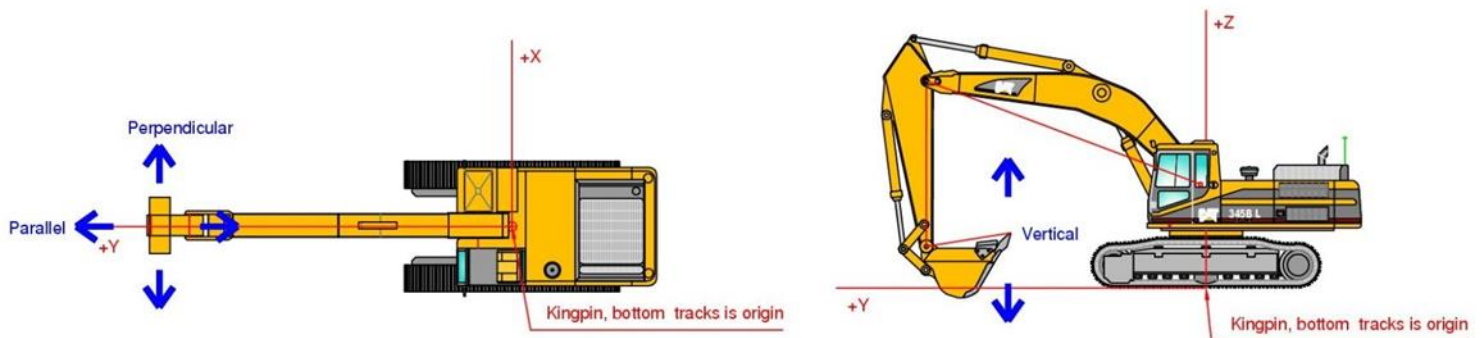


Figure 2: The 3 error components

4.5.1. Horizontal error crane fully extended

Below follows the standard deviation as per the situation where the Boom and stick completely extended.

Ref	Excavator part	Distance / length	Dimension
1	Distance Antenna - Trunnion	2.5	Meters
2	Boom length:	15	Meters
3	Stick length:	10	Meters
4	Bucket size	1.5	Meters
5	Distance between GNSS antenna and Bucket	29	Meters

Ref	Type error	Equipment	Specifications
1	Positioning system error	Trimble SPS85x RTK	0.01m + 0.5ppm
2	Heading error	Trimble SPS55xH	0.1 degr.
3	Roll / Pitch error	XB2i	0.1 degr.
4	Angle measurement devices	Rotax	0.3 degr.

Horizontal error parallel to crane			
Ref	Type error	Error size	Dimension
S ₁	Positioning system	0.01	Meters

S ₂	Roll / Pitch error	0.01	Meters
S ₃	Angle measurement Boom	0.08	Meters
S ₄	Angle measurement Stick	0.05	Meters
S ₅	Angle measurement Bucket	0.01	Meters
Total Standard deviation: $\sqrt{S_1^2 + S_2^2 + S_3^2 + S_4^2 + S_5^2}$		0.096	Meters

Horizontal error perpendicular to crane			
Ref	Type error	Error size	Dimension
P ₁	Positioning system	0.01	Meters
P ₂	Roll / Pitch error	0.01	Meters
P ₃	Heading	0.05	Meters
Total Standard deviation: $\sqrt{P_1^2 + P_2^2 + P_3^2}$		0.052	Meters

4.5.2. Horizontal error crane not extended

Below follows the standard deviation as per the situation where the Boom and stick are not extended.

Ref	Excavator part	Distance / length	Dimension
1	Distance Antenna - Trunnion	2.5	Meters
2	Boom length:	15	Meters
3	Stick length:	10	Meters
4	Bucket size	1.5	Meters
5	Distance between GNSS antenna and Bucket	10	Meters

Ref	Type error	Equipment	Specifications
1	Positioning system error	Trimble SPS85x RTK	0.01m + 0.5ppm
2	Heading error	Trimble SPS55xH	0.1 degr.
3	Roll / Pitch error	Seatools XB2i	0.1 degr.
4	Angle measurement devices	Seatools Rotax HD Digital	0.3 degr.

Horizontal error parallel to crane			
Ref	Type error	Error size	Dimension
S ₁	Positioning system	0.01	Meters
S ₂	Roll / Pitch error	0.01	Meters
S ₃	Angle measurement Boom	0.08	Meters
S ₄	Angle measurement Stick	0.05	Meters

S ₅	Angle measurement Bucket	0.01	Meters
Total Standard deviation: $\sqrt{S_1^2 + S_2^2 + S_3^2 + S_4^2 + S_5^2}$		0.096	Meters

Horizontal error perpendicular to crane			
Ref	Type error	Error size	Dimension
P ₁	Positioning system	0.01	Meters
P ₂	Roll / Pitch error	0.01	Meters
P ₃	Heading	0.02	Meters
Total Standard deviation: $\sqrt{P_1^2 + P_2^2 + P_3^2}$		0.024	Meters

4.5.3. Vertical error

Below follows the standard deviation as per the situation where the boom and stick are completely extended.

Ref	Excavator part	Distance / length	Dimension
1	Distance Antenna - Trunnion	2.5	Meters
2	Boom length:	15	Meters
3	Stick length:	10	Meters
4	Bucket size	1.5	Meters
5	Distance between GNSS antenna and Bucket	29	Meters

Ref	Type error	Equipment	Specifications
1	Positioning system error	Trimble SPS85x RTK	0.02m + 0.5ppm
2	Roll / Pitch error	Seatools XB2i	0.1 degr.
3	Angle measurement devices	Seatools Rotax HD Digital	0.3 degr.

Horizontal error parallel to crane			
Ref	Type error	Error size	Dimension
V ₁	Positioning system	0.02	Meters
V ₂	Roll / Pitch error	0.01	Meters
V ₃	Angle measurement Boom	0.08	Meters
V ₄	Angle measurement Stick	0.05	Meters
V ₅	Angle measurement Bucket	0.01	Meters
Total Standard deviation: $\sqrt{V_1^2 + V_2^2 + V_3^2 + V_4^2 + V_5^2}$		0.097	Meters

5. SURVEY WITH CRANE MONITORING SYSTEM

<<Describe here the survey (related) operations for the project.>>

All surveys will be carried out along pre-determined specifications to be mutually agreed upon between CONTRACTOR and COMPANY.

5.1. Checks prior to survey

Position check will be in accordance with Boskalis procedure "SURV-221 Field Check Position check CMS"

Before an excavator survey can be carried out, checks will be performed, to make sure that the survey data collected by the excavator is correct. Therefore, the land surveyor will place the antenna pole of his RTK GNSS positioning system on top of an official known Benchmark or Survey Control Point and annotates the known- and observed co-ordinates (E, N, Elevation) on a data sheet.

Next the excavator will place the teeth or edge of the bucket on a specially designed measuring plate and co-ordinates will be observed and the location marked as a temporary benchmark. The land surveyor will thereafter hold the antenna pole on the same marked location (temporary benchmark) and both sets of observed co-ordinates will be annotated on the same data sheet.

When the found results are within the horizontal and vertical tolerances, the excavator is ready for the actual excavator bucket survey.

5.2. Acceptance excavator bucket survey

To be able to perform the survey in a proper way, all lines to be surveyed are shown on the Survey computer. These survey lines are pre-defined and are the same as to be used during a bathymetric survey on a survey launch. The lines are perpendicular to the defined centre-line of the work area and have a relative interval of 10m. There are two methods to perform the survey:

- Chainage / Offset;
- Pre-defined dump pattern sets.

5.2.1. Chainage / offset system

The bucket of the excavator will be positioned at the required Chainage value and Offset to the Centre-line. The Chainage and Offset of the bucket are visible on the Crane Monitoring System. Then the position of the bucket (knife or teeth) will be recorded either in Easting, Northing and elevation or Chainage, Offset and elevation. Note that the teeth should be parallel to slopes in order to avoid penetration of the teeth in the slope for applicable situations. Next, the bucket will be moved 1m along the Chainage towards the Centre-line to the next Offset. This procedure will be repeated until the offsets in reach of the excavator on that chainage are surveyed. The excavator will move 10m along the Chainage line and the process is repeated again.

5.2.2. Pre-defined pattern sets

Prior to the survey pre-defined pattern sets have been prepared indicating where survey records will be taken. Pattern sets are a predefined grid. The bucket of the excavator will be positioned in the center of the squares of the grid where a logging will be taken. The square of the pattern set is marked as recorded and the bucket is positioned in the next square. The pattern set will be defined as agreed upon between CONTRACTOR and COMPANY.

5.3. Recordings

The recording of the data will be done in a digital way by the CMS log function which will record a comma delimited file with as a minimum date, time, chainage, offset, easting, northing and elevation.

During the survey, on the CMS screen will be shown which lines are surveyed. When all lines have been surveyed, the data will be further processed in the survey office, so that the survey data can be

presented in cross-profiles or in any other form as agreed upon and be presented for acceptance to COMPANY.

The data handed over to COMPANY shall include, as a minimum Date, Time, Chainage, Offset, Easting, Northing, Elevation, Area, and Layer.

6. REFERENCES, ABBREVIATIONS, DEFINITIONS

6.1. References

Client Documents		
No.	Document No.	Document Title
.....		
.....		

Boskalis Documents	
Document No.	Document Title
GT-256	Project Survey Management Plan
SURV-100	Survey Method Statement
SURV-100b	Definitions & Conditions

Standards	
Document No.	Document Title
ISO 9001:2015	Quality management system
IHO-S44	IHO Standards for hydrographic surveys

6.2. Abbreviations

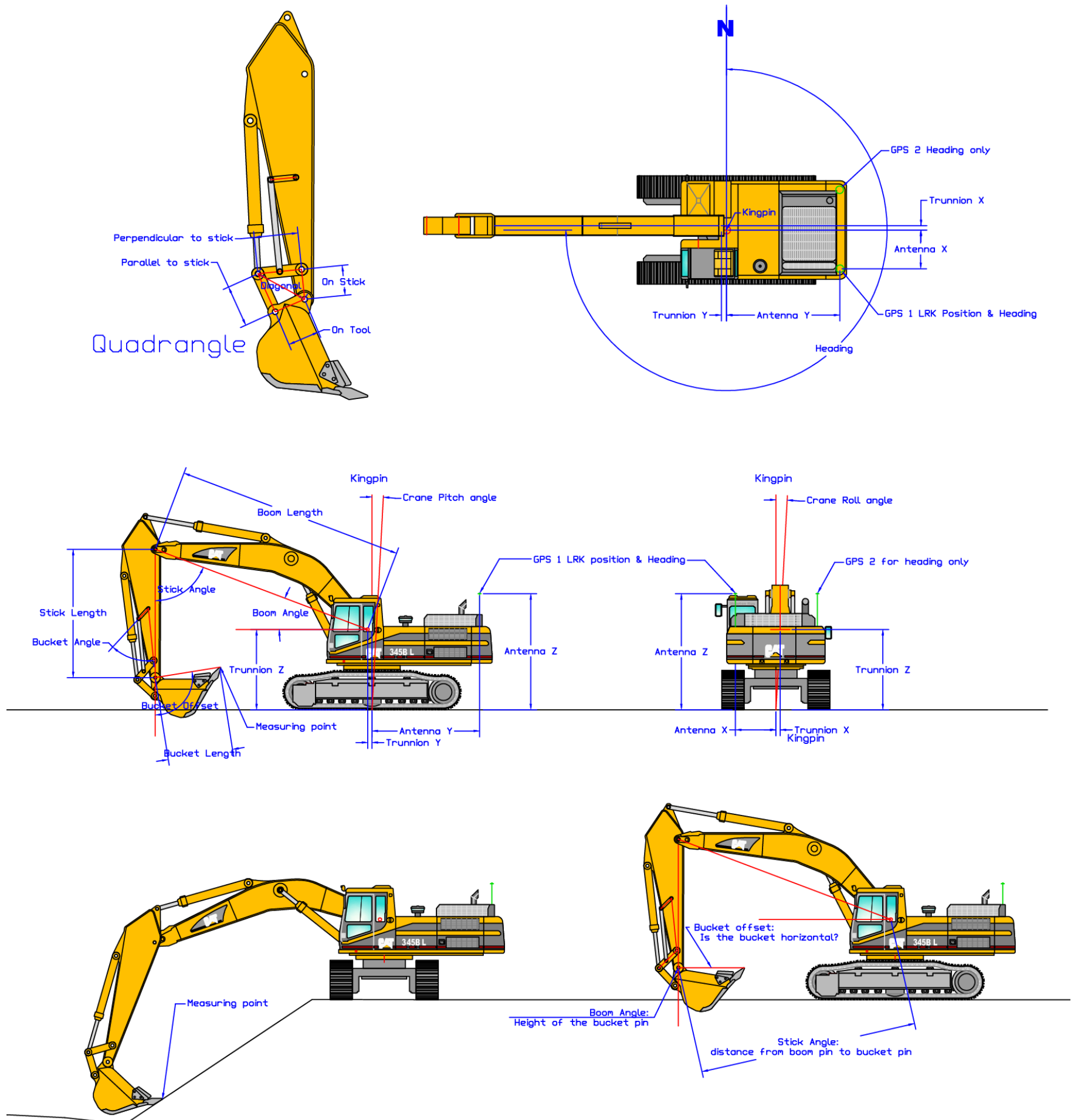
Abbreviation	Full meaning
CMS	Crane Monitoring System
CSV	Comma Separated Values
DGNSS	Differential GNSS
DV	DredgeView Boskalis in-house software
GNSS	Global Navigation Satellite System
HD	Heavy Duty
RMS	Root Mean Square
RTK	Real Time Kinematic
WoW	Way of Working

6.3. Definitions

Definition	Full meaning
COMPANY	Company name
CONTRACTOR	Boskalis
SUBCONTRACTOR	SubContractor name

7. ATTACHMENTS

7.1. CMS offset overview



7.2. Trimble SPS85x RTK GNSS receiver

Specifications		Trimble SPS855 GNSS Modular Receiver	
			
Receiver Name		SPS855 GNSS Modular Receiver	
Configuration Option		Yes, upgradeable to Rover, Base or Rover / Base	
Base and Rover interchangeability		1 Hz, 2 Hz, 5 Hz, 10 Hz, 20 Hz	
Rover position update rate		Unrestricted, typical range 2–5 km (1.2–3 miles) without radio repeater	
Rover maximum range from base radio		Yes	
Rover operation within a VRS™ network		Yes - option ⁷	
Heading and Moving Base operation		See Receiver Upgrades below	
Factory options			
General			
Keyboard and display		Vacuum Fluorescent display 16 characters by 2 rows. Invertable	
		On/Off key for one-button startup	
		Escape and Enter keys for menu navigation	
		4 arrow keys (up, down, left, right) for option scrolls and data entry	
Dimensions (L x W x D)		24 cm x 12 cm x 5 cm (9.4 in x 4.7 in x 1.9 in) including connectors	
Weight		1.65 kg (3.64 lb) receiver with internal battery and radio	
		1.55 kg (3.42 lb) receiver with internal battery and no radio	
Antenna Options			
GA510		L1/L2/L2C GPS, SBAS, and OmniSTAR	
GA530		L1/L2/L2C GPS, SBAS, and OmniSTAR	
GA810		GPS, Glonass, Galileo, Compass, OmniSTAR, SBAS (optimized for OmniSTAR)	
L1/Beacon, DSM 232		Not Supported	
Zephyr™ Model 2		L1/L2/L2C/L5 GPS, Glonass, Galileo, Compass, OmniSTAR, SBAS	
Zephyr Geodetic™ Model 2		L1/L2/L2C/L5 GPS, Glonass, Galileo, Compass, OmniSTAR, SBAS	
Zephyr Model 2 Rugged		L1/L2/L2C/L5 GPS, Glonass, Galileo, Compass, OmniSTAR, SBAS	
Zephyr, Zephyr Geodetic, Z-Plus, Micro-Centered™		Refer to Antenna specification	
Temperature			
Operating ¹		-40 °C to +65 °C (-40 °F to +149 °F)	
Storage		-40 °C to +80 °C (-40 °F to +176 °F)	
Humidity		MIL-STD 810F, Method 507.4	
Waterproof		IP67 for submersion to depth of 1 m (3.3 ft), dustproof	
Shock and Vibration			
Pole drop		Designed to survive a 1 m (3.3 ft) pole drop onto a hard surface	
Shock – Non-operating		To 75 g, 6 ms	
Shock – Operating		To 40 g, 10 ms, saw-tooth	
Vibration		Tested to Trimble ATV profile (4.5 g RMS): 10 Hz to 300 Hz: 0.04 g/Hz ²	
		300 Hz to 1,000 Hz; –6 dB/octave	



Specifications

Trimble SPS855 GNSS Modular Receiver

Measurements	Advanced Trimble Maxwell™ 6 Custom GPS Chips High-precision multiple correlator for GNSS pseudorange measurements Unfiltered, unsmoothed pseudo-range measurements data for low noise, low multipath error, low-time domain correlation, and high-dynamic response Very low noise carrier phase measurements with <1 mm precision in a 1 Hz bandwidth Trimble EVEREST™ multipath signal rejection L-Band: OmniSTAR VBS, HP and XP by subscription GPS L1 C/A, L2C, L2E (Trimble method for tracking unencrypted L2P) upgradable to L5. 440 channels Upgradeable to GLONASS L1/L2C/A, L1/L2P Full Cycle Carrier Upgradeable to Galileo: L1 CBOC, E5A, E5B & E5AltBOC⁸ Upgradeable to Compass: B1, B2, B3 4-channel SBAS L1 C/A, L5 (WAAS/EGNOS/MSAS) QZSS: L1 C/A, L1C, L1 SAIF, L2C, L5
SBAS (WAAS/EGNOS/MSAS) Positioning³	
Accuracy	Better than 5 m 3DRMS (16 ft)
Code Differential GPS Positioning²	
Horizontal accuracy	0.25 m + 1 ppm RMS (0.8 ft + 1 ppm RMS)
Vertical accuracy	0.50 m + 1 ppm RMS (1.6 ft + 1 ppm RMS)
OmniSTAR Positioning	
VBS service accuracy	Horizontal <1 m (3.3 ft)
XP service accuracy	Horizontal 0.2 m (0.66 ft), Vertical 0.3 m (1.0 ft)
HP service accuracy	Horizontal 0.1 m (0.33 ft), Vertical 0.15 m (0.5 ft)
Location RTK Positioning	
Horizontal accuracy	Location RTK (10/10) or (10/2) 10 cm + 1 ppm RMS (0.32 ft + 1 ppm)
Vertical accuracy	Location RTK (10/10) 10 cm + 1 ppm RMS (0.32 ft + 1 ppm)
	Location RTK (10/2) 2 cm + 1 ppm RMS (0.065 ft + 1 ppm)
Real-Time Kinematic (RTK up to 30 km) Positioning²	
Horizontal accuracy	8 mm + 1 ppm RMS (0.026 ft + 1 ppm RMS)
Vertical accuracy	15 mm + 1 ppm RMS (0.05 ft + 1 ppm RMS)
Trimble VRS⁹	
Horizontal accuracy	8 mm + 0.5 ppm RMS (0.026 ft + 0.5 ppm)
Vertical accuracy	15 mm + 0.5 ppm RMS (0.05 ft + 0.5 ppm)
Precise Heading	
Heading accuracy	When combined with SPS555H ⁷
2 m antenna separation	0.09° RMS
10 m antenna separation	0.05° RMS
Initialization Time	
Regular RTK operation with base station	Single/Multi-base
	typically less than 8 seconds
Initialization reliability ⁴	>99.9%
Power	
Internal	Integrated internal battery 7.2 V, 7800 mA-hr, Lithium-ion
	Internal battery operates as a UPS in the event of external power source failure
	Internal battery will charge from external power source as long as source can support the power drain
	Integrated charging circuitry



7.3. Trimble SPS55xH GNSS Heading add-on

Specifications		Trimble SPS555H Heading Add-on Receiver	
			
Receiver Name		SPS555H Heading Add-on receiver	
Configuration Option			
Base and Rover interchangeability			N/A
Rover position update rate		1 Hz, 2 Hz, 5 Hz, 10 Hz, 20 Hz	
Rover maximum range from base radio		2400m	
Rover operation within a VRS™ network			N/A
Heading and Moving Base operation			Heading only
Factory options		See Receiver Upgrades below	
General			
Keyboard and display		Vacuum Fluorescent display 16 characters by 2 rows. Invertable On/Off key for one-button startup Escape and Enter keys for menu navigation 4 arrow keys (up, down, left, right) for option scrolls and data entry	
Dimensions (L x W x D)		24 cm x 12 cm x 5 cm (9.4 in x 4.7 in x 1.9 in) including connectors	
Weight			N/A
		1.55 kg (3.42 lb) receiver with internal battery and no radio	
Antenna Options			
GA510			Not included in a kit
GA530			Not included in a kit
GA810		GPS, Glonass, Galileo, Compass, SBAS. Included in standard kit	
L1/Beacon, DSM 232			Not Supported
Zephyr™ Model 2		L1/L2/L2C/L5 GPS, Glonass, Galileo, Compass, OmniSTAR, SBAS	
Zephyr Geodetic™ Model 2		GPS, Glonass, Galileo, Compass, SBAS. Included in Precise kit	
Zephyr Model 2 Rugged		GPS, Glonass, Galileo, Compass, SBAS. Included in Rugged kit	
Zephyr, Zephyr Geodetic, Z-Plus, Micro-Centered™		Refer to Antenna specification	
Temperature			
Operating ¹		-40 °C to +65 °C (-40 °F to +149 °F)	
Storage		-40 °C to +80 °C (-40 °F to +176 °F)	
Humidity		MIL-STD 810F, Method 507.4	
Waterproof		IP67 for submersion to depth of 1 m (3.3 ft), dustproof	
Shock and Vibration			
Pole drop		Designed to survive a 1 m (3.3 ft) pole drop onto a hard surface	
Shock – Non-operating		To 75 g, 6 ms	
Shock – Operating		To 40 g, 10 ms, saw-tooth	
Vibration		Tested to Trimble ATV profile (4.5 g RMS): 10 Hz to 300 Hz: 0.04 g/Hz ² 300 Hz to 1,000 Hz; –6 dB/octave	



Specifications	Trimble SPS555H Heading Add-on Receiver
Measurements	Advanced Trimble Maxwell™ 6 Custom GPS Chips High-precision multiple correlator for GNSS pseudorange measurements Unfiltered, unsmoothed pseudo-range measurements data for low noise, low multipath error, low-time domain correlation, and high-dynamic response Very low noise carrier phase measurements with <1 mm precision in a 1 Hz bandwidth Trimble EVEREST™ multipath signal rejection GPS L1 C/A, L2C, L2E (Trimble method for tracking unencrypted L2P) upgradable to L5. 440 channels Upgradeable to GLONASS L1/L2C/A, L1/L2P Full Cycle Carrier Upgradeable to Galileo: L1 CBOC, E5A, E5B & E5AltBOC⁸ Upgradeable to Compass: B1, B2, B3 4-channel SBAS L1 C/A, L5 (WAAS/EGNOS/MSAS) QZSS: L1 C/A, L1C, L1 SAIF, L2C, L5
SBAS (WAAS/EGNOS/MSAS) Positioning³	
Accuracy	N/A
Code Differential GPS Positioning²	
Horizontal accuracy	N/A
Vertical accuracy	
OmniSTAR Positioning	
VBS service accuracy	N/A
XP service accuracy	
HP service accuracy	
Location RTK Positioning	
Horizontal accuracy	N/A
Vertical accuracy	
Real-Time Kinematic (RTK up to 30 km) Positioning²	
Horizontal accuracy	N/A
Vertical accuracy	
Trimble VRS⁹	
Horizontal accuracy	N/A
Vertical accuracy	
Precise Heading	
Heading accuracy	When combined with SPS855H ⁷
2 m antenna separation	0.09° RMS
10 m antenna separation	0.05° RMS
Initialization Time	
Regular RTK operation with base station	N/A
Initialization reliability ⁴	
Power	
Internal	Integrated internal battery 7.2 V, 7800 mA-hr, Lithium-ion Internal battery operates as a UPS in the event of external power source failure Internal battery will charge from external power source as long as source can support the power drain Integrated charging circuitry



7.4. AE Sensors XB2i

seika.de®

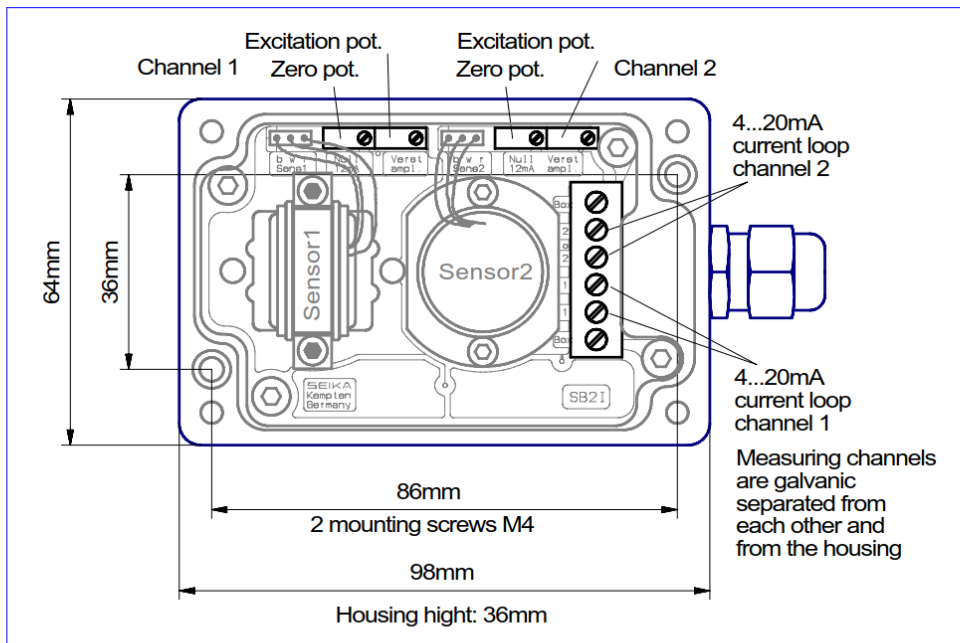
SB2i

Technical Specifications

Terminals	6 x 1,5 mm ²
Cable fixing	M12 x 1.5 cable gland, clamping range 6mm ... 7.5mm
Measuring range, Resolution, etc.	depends on implemented SEIKA sensors
Degree of protection	IP65
Mounting orientation	any
Measuring planes (N... sensors)	3 main housing planes
Measuring directions (B..., BD... sensors)	In X,Y,Z coordinate of housing
Terminal voltage	10V ... 30V
Minimum loop currents	2.5mA ... 3.5mA
Maximum loop currents	22mA ... 26mA
Output signal loop current	4mA ... 20mA (12mA for zero position)
Adjustable variables	zero point (12mA), amplification
Maximum load resistances	500_Ohm (at 24 Volt supply voltage)
Low pass filter	Active, 3rd order, minimal ripple
Operating temperature	-40°C ... +85°C

Options: special measuring ranges, calibration record, silicon encapsulation, custom wiring

Dimensions (in mm)



SEIKA Mikrosystemtechnik GmbH - Ellharter Str.10 - D-87435 Kempten - Tel: 0831-25532 Fax: 0831-25534

Internet: <http://www.seika.de> - <http://www.seika.net> - Email: seika@seika.de



XB...
Inclinometer

XB...

Revised model

The XB housing is an extra-robust, pressure-, saltwater- and corrosion-proof stainless steel housing for one or two SEIKA sensors. It can contain the same sensor models and electronics like the SB-housing. It is available with a compact stainless steel cable gland and a Subconn cable gland for seawater and underwater usage.



Features

- extra robust, pressure-, saltwater- and corrosion-proof stainless steel housing
- twist-free 4-point fastening of rigid 3.2mm thick base PCB
- can be equipped with one or two SEIKA sensors, like the SB-housing
- all SEIKA sensors fit the housing and can be installed in different directions of operation
- sensors and sensor electronics electrically isolated from housing
- individual calibration protocol
- for electrical characteristics, please see the datasheets of the corresponding SB-sensorboxes, like SB11, SB2I, SB1S, SB1U, SBS1U and SB360

Description

The XB... is an extra robust, pressure-, saltwater- and corrosion-proof stainless steel housing with an integrated sensor or sensors and one of the sensor electronic systems described for the SB... sensor casings. The compact stainless steel cable gland and stainless steel housing size enable the use of this high quality measuring system in especially harsh operating conditions.

Application

The XB... is suited for taking precise inclination or acceleration measurements under harsh circumstances and returning of a 0...5V or 4...20mA output signal. Areas of application include construction, mining, agricultural machinery, transportation and conveyor systems, ships, in aggressive fluids and in saltwater, in high pressure environments, operation and automation technology as well as general mechanical engineering.

WWW.AESENSORS.NL

7.5. Seatools Rotax HD



seatools



UNDERWATER SPECIALISTS

SELECTION & OPTION LIST

For every Rotax® HD Digital the following configuration items must be determined:

- **Rotation angle range.** The sensor can be supplied in either a single-turn or multi-turn version.
- **Data output format.** The Rotax® HD Digital can provide the following data output formats: SSI, CAN bus, Profibus, RS 422 / 485, and EtherCAT.
- **Connector type.** The sensor is typically supplied with Subconn® connectors. Other connector types are available upon request.

The following features are optional:

A Shaft clamp

This shaft clamp, to be bolted onto the sensor shaft, can be used to connect (via a welded connection) to a lever to establish a mechanical connection to, for example, a crane boom.

B Cable assembly

The sensor can be supplied with custom-made cabling in a variety of lengths, allowable loads, and connector types.



Subconn® Micro 6-pole Male



TYPICAL APPLICATIONS

The Rotax® HD Digital subsea rotation sensor is specifically designed to perform accurate rotation measurement. The sensor can be installed on equipment operating in harsh subsea environments, including:

- Dredging equipment such as booms, bucket angles, and suction tubes
- Offshore structures such as A-frames
- Heavy-duty cranes
- Winches



ROTAX® HD DIGITAL SUBSEA ROTATION SENSOR SPECIFICATIONS

GENERAL

	METRIC	IMPERIAL
Main dimensions	See detailed drawings	
Material housing / shaft	Stainless steel 316L / 431	
Weight in air	± 12.5 kg	± 27.6 lbs
Maximum static axial shaft load	5 kN	1124 lbf
Maximum static radial shaft load	6 kN	1349 lbf
Maximum working depth	100 m	328 ft
Operational temperature range	-10°C - +60 °C	14 °F - +140 °F

ELECTRICAL – PROFIBUS DATA OUTPUT

Supply voltage	10 - 30 V DC
Current consumption	230 mA (max.)
Maximum resolution	Single turn: 16 bits, Multi turn: 14 bits
Output code	Profibus DP
Connector	Subconn® Micro 6-pole Male
Transmission rate	0,0096 - 12 Mbit/s

Electrical specifications of HD Digital sensors executed with other data output formats are available upon request.

7.6. DV Crane



INTRODUCTION

A shore-based hydraulic excavator with position monitoring can accurately perform various tasks at locations that would otherwise be inaccessible because of the excavator's tracks. These tasks are mainly performed onshore or in the transition between land and water. The excavator can also operate from a pontoon and thus practically becomes a backhoe dredger. The positioning system enables the excavator to position the bucket with an accuracy of a few centimeters.

EXCAVATOR WITH CRANE MONITORING SYSTEM (CMS)

The key parts of an excavator with positioning are:

- Excavator on tracks with boom stick and bucket, grab or another tool
- Positioning system consisting of high accuracy Real Time Kinematic Global Navigation Satellite System (RTK GNSS), GNSS heading sensor, Roll/pitch sensor, angler sensors
- Boskalis DredgeView Crane Monitoring System (DV Crane)

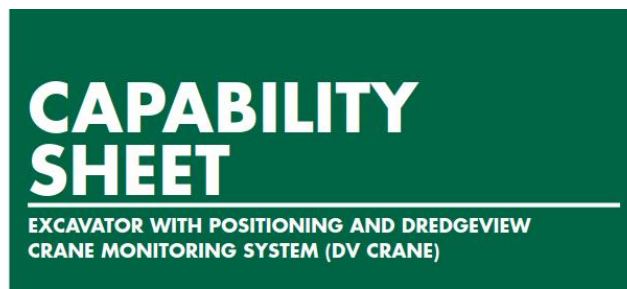
SUITABILITY

Excavators with positioning are used in dredging and backfill works for shore approaches, shore protection and roadworks. In dredging works, the excavator can dredge near shore within reach of the crane on difficult to reach locations. In shore protection works, the excavator is mainly used to place geotextile, graded filter stone, rock and Xblocs, and in road works to profile sand material.

WORK METHOD

For dredging, the excavator can be placed on elevated tracks, on a pontoon, or can work from the shore. The Crane Monitoring System (CMS) and positioning show the position of the bucket opposite to any required dredging design.

In shore protection works, the excavator is used to profile the under layers, to place the geotextile, the



- A Excavator with CMS
- B Excavator Dredging and profiling with DV Crane
- C Stick Angler

filter layers and stones, and finally to place the concrete top layer of Xblocs, accropodes, or similar material. The different layers are visualized by the CMS positioning system to enable the operator to place each layer to the required thickness. The grid shown in the top view enables the operator to place the Xblocs in the defined grid, and in the defined orientation when the echoscope is used.

DV CRANE

Boskalis developed DV Crane, an advanced system for positioning excavators and backhoes. The system displays the excavator and its bucket, grab, or any other configuration together with design levels and survey levels in a top view, side view and 3D view. With input from the accurate positioning, heading, roll/pitch and angler sensors, the accuracy of the measuring point can be achieved to within 10 cm in a horizontal and vertical plane.

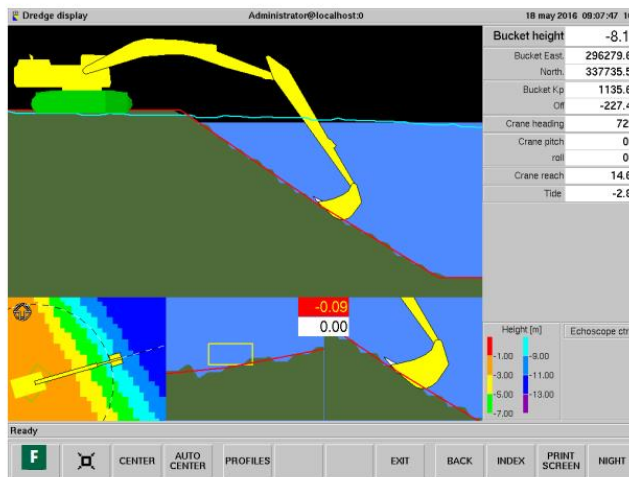
HARDWARE USED WITH DV CRANE

The following hardware is used for a standard DV Crane setup:

- Trimble SPS85x RTK GNSS receiver
- Trimble SPS55xH Heading RTK GNSS receiver
- AE sensors XB2i roll and pitch sensor
- Seatools Rotax HD digital Angler sensors
- Nexcom fanless industrial computer NISE with DV Crane software.



EXCAVATOR WITH POSITIONING AND DREDGEVIEW CRANE MONITORING SYSTEM (DV CRANE)



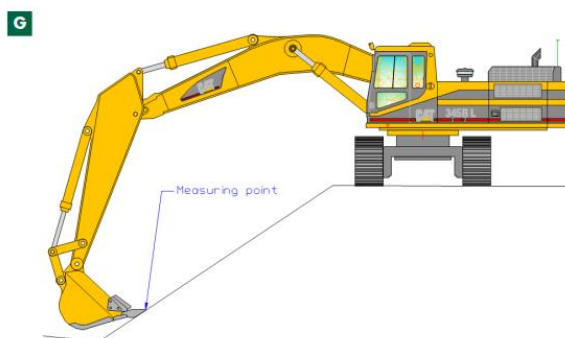
TOPOGRAPHIC AND HYDROGRAPHIC SURVEYS

The high accuracy of the positioning system enables surveys to be performed at locations where other survey systems cannot be used, for instance, for safety reasons. On stone slopes or at the transition between land and hydrographic surveys, survey points can be taken within reach of the bucket or tool. The data are saved in a comma delimited file. The logged files are processed in the Boskalis DV-Office Survey package and converted to a 3D digital terrain model (DTM) for use in producing charts, cross profiles and volume calculations. The data can be used as final as-built survey.

ECHOSCOPE

Boskalis has the option to install the echoscope for placing Xblocs, Accropodes and other concrete blocks for shore protection. This is a complicated process in which the Xblocs are placed in an interlocking pattern in a specified orientation.

While the DV Crane is used to position an Xbloc, the echoscope provides an accurate 3D view in real time of the Xbloc underwater in relation to the adjacent Xblocs. The tool at the end of the excavator's stick is used to rotate the Xbloc to place it accurately underwater and in the correct orientation.



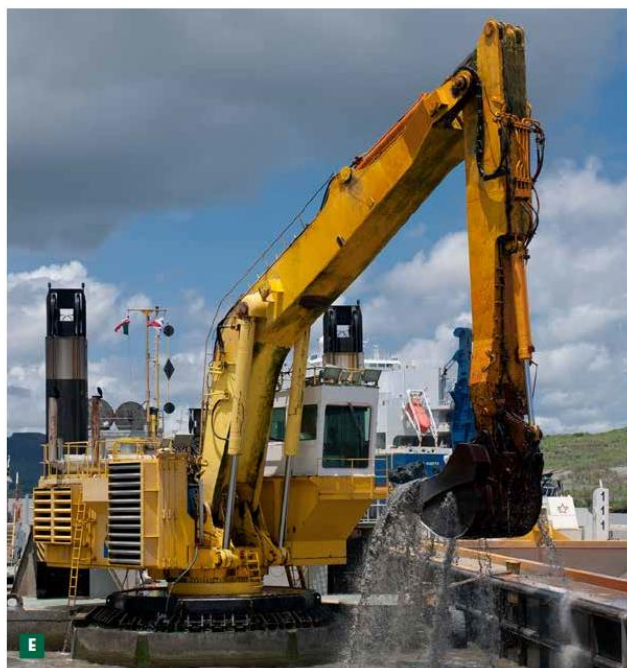
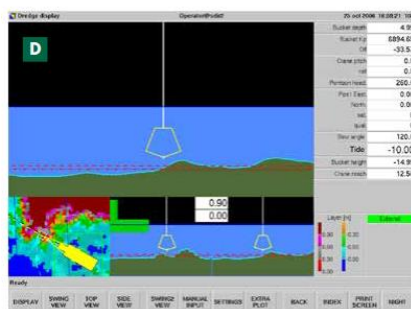
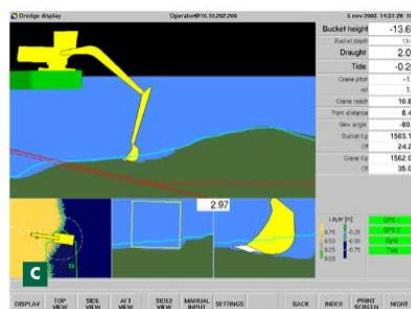
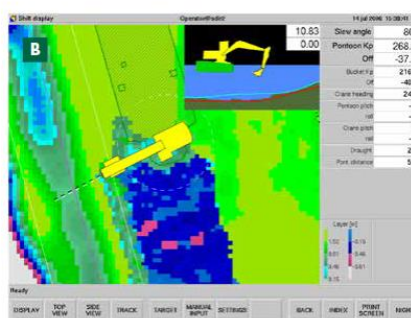
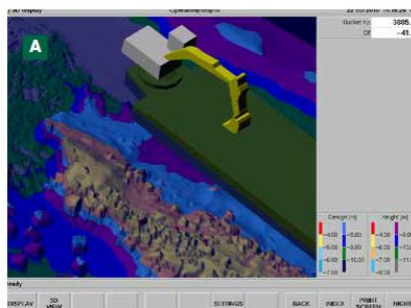
- D Echoscope installed on XBloc installation tool
- E Echscope and DV Crane displays
- F Typical DV Crane screen pages
- G Survey with CMS of excavator system

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CAPABILITY SHEET

DV CRANE

BACKHOE AND GRAB DREDGER POSITIONING & PRODUCTION MODULE



BOSKALIS DV SOFTWARE SUITE

DV Crane is the Boskalis DV Software Suite backhoe & grab dredger positioning and production module, custom tuned to each Boskalis backhoe or grab dredger.

DV Crane integrates precise 3D positioning, dredger modelling, project design information and survey results into real time graphical displays for both crane and barge operators. The provided graphical displays are a plan view, a profile view and a 3D view.

These displays allow the operators to dredge or build accurately to the required design. The position of the grab or bucket can be logged in readable ASCII format at time interval, at deepest point of excavation, or in case of backfill at the position of each individual dump.

For dredging near existing infrastructure like pipelines or cables the software provides the option to display the infrastructure, and has built in proximity alarms to ensure safety for crane, personnel and the infrastructure.

- A DV Crane 3D display
- B DV Crane Shift Display
- C DV Crane Dredge Display
- D DV Crane Wire Dredge Display
- E Boskalis Backhoe Dredger

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